Prayas JEE AIR (2025)

WEEKLY TEST 25

DURATION : 180 Minutes | **DATE : 29/12/2024** | **M. MARKS : 300**

ANSWER KEY

PHYSICS

1. (4)
2. (4)
3. (3)
4. (3)
5. (3)
6. (2)
7. (1)
8. (3)
9. (3)
10. (4)
11. (1)
12. (4)
13. (2)
14. (3)
15. (4)
16. (3)
17. (3)
18. (4)
19. (2)
20. (1)
21. (35)
22. (200)
23. (120)
24. (1)
25. (20)

CHEMISTRY

26. (2)
27. (2)
28. (2)
29. (2)
30. (4)
31. (1)
32. (2)
33. (2)
34. (3)
35. (4)
36. (1)
37. (3)
38. (2)
39. (4)
40. (4)
41. (4)
42. (1)
43. (1)
44. (3)
45. (2)
46. (15)
47. (128)
48. (25)
49. (4)
50. (25)

MATHEMATICS

51. (4)
52. (3)
53. (2)
54. (1)
55. (3)
56. (4)
57. (1)
58. (4)
59. (4)
60. (3)
61. (2)
62. (2)
63. (2)
64. (3)
65. (3)
66. (3)
67. (1)
68. (3)
69. (3)
70. (4)
71. (7)
72. (2)
73. (18)
74. (36)
75. (120)

SECTION-I (PHYSICS)

1. **(4)**

(*d*) Time to reach at max height, $t = \frac{u}{g}$

No. of balls thrown in 1 sec = *n*.

So, time taken by each ball $\frac{1}{n}$ sec

i.e. $\frac{u}{g} = \frac{1}{n} \Rightarrow u = \frac{g}{n}$

So, $h_{\max} = \frac{u^2}{2g} = \frac{g^2}{2gn^2} = \frac{g}{2n^2}$

2. **(4)**

$\because \frac{Q^2}{4\pi\varepsilon_0 R_0} = \frac{Q^2}{4\pi\varepsilon_0 R} + \frac{1}{2}mv^2$

$\Rightarrow \mathrm{v} = \sqrt{\frac{Q^2}{4\pi\varepsilon_0} \times \frac{2}{m}\left[\frac{1}{R_0} - \frac{1}{R}\right]}$

So *v* increases and attains a finite value after large time.

3. **(3)**

Angular momentum conserved about *O*

$mvl = \left(\frac{Ml^2}{3} + ml^2\right)\omega \Rightarrow \omega = \frac{3mv}{3ml + Ml} = \frac{18}{5}\,\mathrm{rad/s}$

Energy conservation

$\frac{1}{2}\left(\frac{MI^2}{3} + ml^2\right)\omega = \left(Mg\frac{I}{2} + mgI\right)(1 - \cos\theta)$

$\Rightarrow \frac{1}{2}\left(\frac{5}{3}\right) \times \frac{324}{25} = (10+10)(1-\cos\theta)$

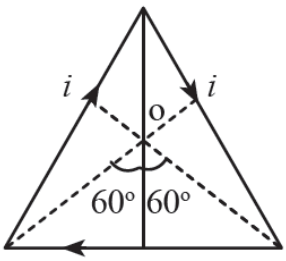
$\Rightarrow 20(1-\cos\theta) = \frac{324}{30}$

$\cos\theta = 1 - \frac{324}{600} = \frac{276}{600} = 0.46 \Leftrightarrow \theta \simeq 63^\circ$

4. **(3)**

$B = \frac{\mu_0 i}{4\pi r}\left(\sin\theta_1 + \sin\theta_2\right)$

$= \frac{10^{-7} \times 1.5}{\left(\frac{0.09}{2\sqrt{3}}\right)} \times \left(\sin 60^\circ + \sin 60^\circ\right) = 10^{-5} T$



Net magnetic field at centroid is $B_{net} = 3\times10^{-5}\,\mathrm{T}$ inside the plane

5. **(3)**

$(a)\, I = \frac{ML^2}{12}$

$(b)\, I = \frac{(2M)L^2}{3}$

$(c)\, I = \frac{M(2L)^2}{12} = \frac{ML^2}{3}$

$(d)\, I = \frac{(2M)(2L)^2}{3} = \frac{8ML^2}{3}$

6. **(2)**

From the formula $n = \frac{\text{real depth}}{\text{apparent depth}}$

$d_{\text{app}} = \frac{d_1}{n_1} + \frac{d_2}{n_2}$

$d_{\text{app}} = \frac{d}{2}\left[\frac{n_1 + n_2}{n_1 n_2}\right]$

7. **(1)**

$\text{Stress} = \frac{\text{Weight}}{\text{Area}}$

$\Rightarrow \frac{10}{2.5\times10^{-4}} = \frac{25}{A}$

$\Rightarrow A = 6.25\times10^{-4}$

8. **(3)**

EM wave is in direction $\rightarrow \frac{\hat{i} + \hat{j}}{\sqrt{2}}$

Electric field is in direction $\rightarrow \hat{k}$

$E' \times B' \rightarrow$ direction of propagation of EM wave.

9. **(3)** Pressure decreases with increase in volume, in both isothermal and adiabatic process. In adiabatic process, volume decreases and pressure increases with increase in temperature.

10. **(4)**

Given: $m(Li) = 7.0160$ u

$m(He) = 4.0026\ u$

$m(H) = 1.0079\ u$

${}^7_3Li + {}^1_1H \rightarrow 2{}^4_2\mathrm{He}$

$\Delta m = \left(m_{\mathrm{Li}} + m_{\mathrm{H}} - 2m_{\mathrm{He}}\right)$

$= .0187u$

Energy liberates for 1gLi $= \frac{\Delta mc^2}{m(Li)}$

Energy liberates for 20 g Li $= \frac{\Delta mc^2}{m(Li)} \times 20$

$$= \frac{0.0187 \times 931.5 \times 10^6 \times 1.6 \times 10^{-19} \times \frac{20}{7} \times 6.023 \times 10^{23}}{36 \times 10^5}$$

$= 1.33 \times 10^6$ kWh

11. (1)

We know that a dipole is a pair of equal and opposite charges separated by a small distance.

$\because$ The net charge on the electric dipole,

$q_{net} = +q - q = 0$

Therefore, according to Gauss's law,

Electric Flux, $\oint \vec{E} \cdot \overrightarrow{ds} = \frac{q_{net}}{\varepsilon_0} = \frac{0}{\varepsilon_0} = 0$

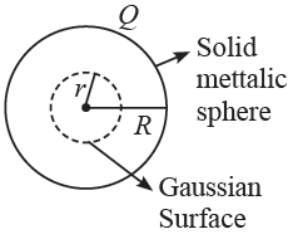
Where, $\vec{E}$ is the electric field, $\overrightarrow{ds}$ is the surface area a small section, and ε_0 is permittivity in free space.

And for a small section $\overrightarrow{ds}$ only, $\vec{E} \cdot \overrightarrow{ds} \neq 0$

$\because$ The flux of the electric field throughout the hollow sphere is zero and the electric field is non-zero.

Hence, statement-I is true :

Electric field in region (r < R)



$\vec{E} = 0$ (inside shell theorm)

$\therefore$ The electric field due to the charged solid sphere at a distance r from the centre is zero. As change encloses within the gaussian surface is equal to zero such as.

$\phi = \oint \vec{E} \cdot \overrightarrow{ds} = 0$

Hence, Statement-II is false:

12. (4)

It is given that, $F = -ax^2$

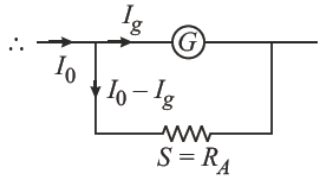
$$\Rightarrow \frac{mvdv}{dx} = -\alpha x^2$$

$$\Rightarrow \int_{v_0}^{0} vdv = -\frac{\alpha}{m}\int_0^x x^2 dx \Rightarrow \frac{0^2}{2} - \frac{v_0^2}{2} = \frac{-\alpha}{3m}x^3$$

$$\Rightarrow \frac{-v_0^2}{2} = -\frac{\alpha}{3m}x^3 \Rightarrow x = \left(\frac{3v_0^2 m}{2\alpha}\right)^{1/3}$$

13. (2)

When galvanometer is used as an ammeter, shunt is used in parallel with galvanometer.



$\therefore I_g G = (I_0 - I_g) R_A$

$\therefore R_A = \left(\frac{I_g}{I_0 - I_g}\right) G$

When galvanometer is used as a voltmeter, resistance is used in series with galvanometer.



$I_g(G + R_V) = V = GI_0$ (given $V = GI_0$)

$\therefore R_V = \frac{(I_0 - I_g)G}{I_g}$

$\therefore R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

14. (3)

Along horizontal direction,

$M \times 10 \cos 30^\circ + 2M \times 5 \cos 45^\circ$

$= 2M \times v_1 \cos 30^\circ + M\, v_2 \cos 45^\circ$

$$\Rightarrow 5\sqrt{3} + 5\sqrt{2} = 2v_1 \frac{\sqrt{3}}{2} + \frac{v_2}{\sqrt{2}}$$

Along vertical direction, $M \times 10 \sin 30^\circ - 2M \times 5 \sin 45^\circ = M\, v2 \sin 45^\circ - 2M\, v_1 \sin 30^\circ$

$$\Rightarrow 5 - 5\sqrt{2}$$

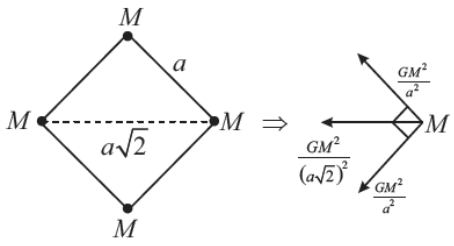
$$= \frac{v_2}{\sqrt{2}} - v_1$$

Solving $v_1 = \frac{17.5}{2.7} = 6.5\,\text{m/s}$ and $v_2 \approx 6.3\,\text{m/s}$

15. (4)

Current direction reverses when magnet exits from solenoid and when magnet is completely inside, current is zero.

16. (3)



Net force on particle towards centre of circle is

$$F_C = \frac{GM^2}{2a^2} + \frac{GM^2}{a^2}\sqrt{2}$$

$$= \frac{GM^2}{a^2}\left(\frac{1}{2} + \sqrt{2}\right)$$

This force will act as centripetal force. Distance of particle from centre of circle is $\frac{a}{\sqrt{2}}$.

$$r = \frac{a}{\sqrt{2}}, F_C = \frac{mv^2}{r}$$

$$\frac{mv^2}{a/\sqrt{2}} = \frac{GM^2}{a^2}\left(\frac{1}{2} + \sqrt{2}\right)$$

$$v^2 = \frac{GM}{a}\left(\frac{1}{2\sqrt{2}} + 1\right)$$

$$v^2 = \frac{GM}{a}(1.35)$$

$$v = 1.16\sqrt{\frac{GM}{a}}$$

17. (3)

Formula of fringe width $(\beta) = \frac{D\lambda}{d} \Rightarrow \beta \propto \lambda$

$$\therefore \Rightarrow \frac{\beta_2}{\beta_1} = \frac{\lambda_2}{\lambda_1} \Rightarrow \beta_2 = \frac{\lambda_2}{\lambda_1} \times \beta_1$$

$$\Rightarrow \beta_2 = \frac{600\,\text{nm}}{400\,\text{nm}} \times 2\,\text{mm}$$

$$\Rightarrow \beta_2 = 3\,\text{mm}$$

18. (4)

$(P_2 - P_1)A = \text{mg}$

$$\Rightarrow P_2 - P_1 = \frac{5.4 \times 10^6}{500}$$

$$\Rightarrow P_2 - P_1 = 10.8 \times 10^3\ \text{N/m}^2$$

Using Bernoulli's theorem,

$$P_2 + 0 + \frac{1}{2}\rho v_2^2 = P_1 + 0 + \frac{1}{2}\rho v_1^2$$

$$P_2 - P_1 = \frac{1}{2}\rho(v_1 - v_2)(v_1 + v_2)$$

$$10.8 \times 10^3 = \frac{1}{2} \times 1.2(v_1 - v_2) \times 2 \times 3 \times 10^2$$

$$\Rightarrow v_1 - v_2 = 30$$

$$\therefore \left(\frac{v_1 - v_2}{v}\right) \times 100 = \frac{30}{300} \times 100 = 10\%$$

19. (2)

P.E. = 0 [For every large distance]

∴ Total energy = 2.6 + 0 = 2.6 eV

In 1st excited state of H, total energy = –3.4 eV

Thus loss in total energy = 6 eV

$$\text{Frequency of emitted photon} = \frac{\text{Energy of photon}}{h}$$

$$= \frac{6 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}} = 1.45 \times 10^{15}\ \text{Hz} = 1.45 \times 10^9\,\text{mHz}$$

20. (1)

As we know that, $v^2 = \omega^2\left(A^2 - x^2\right)$ for SHM

$$\therefore v_1^2 = \omega^2\left(A^2 - x_1^2\right)$$

$$v_2^2 = \omega^2\left(A^2 - x_2^2\right)$$

$$v_1^2 - v_2^2 = \omega^2\left(x_2^2 - x_1^2\right) \Rightarrow \omega - \sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}}$$

$$\Rightarrow T = \frac{2\pi}{\omega} \Rightarrow T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$$

21. (35)

Given,

Denisty of wire, $\sigma = 9 \times 10^{-3}\,\text{kg cm}^{-3}$

Young's modulus of wire, Y = 9×10^{10} Nm^{-2}

Strain = 4.9×10^{-4}

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{T/A}{\text{Strain}}$$

$$\therefore \frac{T}{A} = Y \times \text{ Strain } = 9 \times 10^9 \times 4.9 \times 10^{-4}$$

Also, mass of wire, m = $Al\,\sigma$

Mass per unit length, $\mu = \frac{m}{J} = A\sigma$

Fundamental frequency in the string

$$f = \frac{1}{2l}\sqrt{\frac{T}{\mu}} = \frac{1}{2l}\sqrt{\frac{T}{\sigma A}}$$

$$= \frac{1}{2 \times 1}\sqrt{\frac{9 \times 10^9 \times 4.9 \times 10^{-4}}{9 \times 10^3}}$$

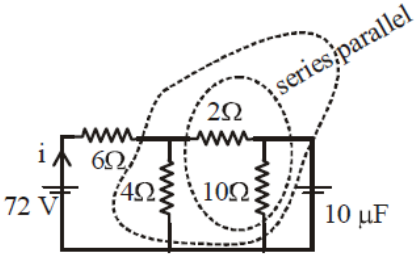
$$= \frac{1}{2}\sqrt{49 \times 10^{9-4-3}} = \frac{1}{2} \times 70 = 35\,\text{Hz}$$

22. (200)

At steady state, there is no current in capacitor.

2Ω and 10Ω are in series. There equivalent resistance is 12Ω. This 12Ω is in parallel with 4Ω and there combined resistance is 12 × 4/(12 + 4). This resistance is in series with 6Ω. Therefore, current drawn from battery

$$i = \frac{V}{R} = \left(\frac{72}{6 + \frac{12 \times 4}{12 + 4}}\right) = 8\,\text{A}$$



Current in 10Ω resistor

$i' = \left(\frac{4}{4+12}\right)8 = 2A$

Pd across capacitor, V = i' R = 2 × 10 = 20V

$\therefore$ Charge on the capacitor, q = CV

= 10 × 20 = 200 μC.

23. (120)

The impedance, $Z = \frac{V_{rms}}{i} = \frac{200}{2} = 100\Omega$

We know that Power, $P = \frac{V_{rms}^2 R}{Z^2}$

or $320 = \frac{200^2 \times R}{100^2}$

$\therefore \quad R = 80\Omega$

We know that $Z = \sqrt{R^2 + X_L^2}$

Or $100 = \sqrt{80^2 + X_L^2}$

$\therefore \quad X_L = 60\Omega$

The back emf, $V_L = iX_L = 2 \times 60 = 120\,\text{V}$

24. (1)

Let ϕ = work function of the metal,

$\frac{hc}{\lambda_1} = \phi + eV_1 \qquad \ldots.(1)$

$\frac{hc}{\lambda_2} = \phi + eV_2 \qquad \ldots.(2)$

Sutracting (ii) from (i) we get

$hc\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right) = e(V_1 - V_2)$

$\Rightarrow V_1 - V_2 = \frac{hc}{e}\left(\frac{\lambda_2 - \lambda_1}{\lambda_1 \cdot \lambda_2}\right)$ $\left[\begin{array}{l}\lambda_1 = 300\,\text{nm} \\ \lambda_2 = 400\,\text{nm} \\ \frac{hc}{e} = 1240\,\text{nm} - \text{V}\end{array}\right]$

$= (1240\,\text{nm} - \text{v})\left(\frac{100\,\text{nm}}{300\,\text{nm} \times 400\,\text{nm}}\right)$

$= 1.03\,\text{V} \approx 1\text{V}$

25. (20)

Volume capacity of beaker, V_0 = 500 cc

$V_b = V_0 + V_0\gamma_{\text{beaker}}\,\Delta T$

When beaker is partially filled with V_m volume of mercury,

$V_m^1 = V_m + V_m\gamma_m\Delta T$

Unfilled volume $(V_0 - V_m) = (V_b - V_m^1)$

$\Rightarrow V_0\gamma_{\text{beaker}} = V_m\gamma_M$

$\therefore V_m = \frac{V_0\gamma_{\text{beaker}}}{\gamma_M}$

or, $V_m = \frac{500 \times 6 \times 10^{-6}}{1.5 \times 10^{-4}} = 20\text{cc}.$

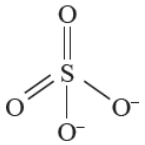
SECTION-II (CHEMISTRY)

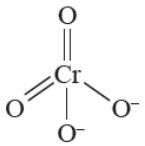
26. (2)

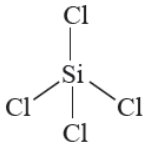
As in H_3PO_4 phosphorus is present in it's maximum oxidation state hence it cannot act as reducing agent.

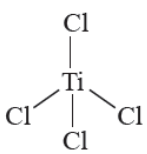
27. (2)

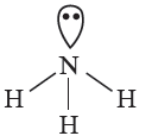
Species having same structures are isostructural.

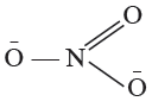
A. SO_4^{2-}  Tetrahedral

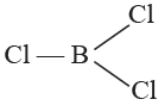
CrO_4^{2-}  Tetrahedral

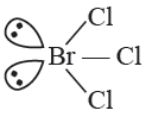
B. $SiCl_4$  Tetrahedral

$TiCl_4$  Tetrahedral

C. NH_3  Pyramidal

NO_3^-  Trigonal planar

D. BCl_3  Trigonal planar

$BrCl_3$  Bent-T

28. (2)

Carbocation B is more stable than carbocation A, so E_a for II step is small, hence B forms faster as compared to A.

29. (2)

$Cu(s) + 2Ag^+(aq) \longrightarrow Cu^{2+}(aq) + 2Ag(s)$

$K_c = 10^{15} \times 10$

$E^o_{cell} = ?$

$E^o_{cell} = \frac{2.303RT}{2F}\log\left(10^{15} \times 10\right)$

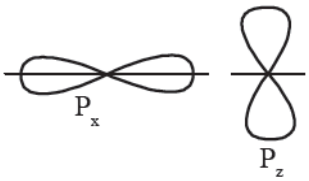
$= \frac{0.059}{2} \times [16\log 10]$ [log 10 = 1]

$E^\circ_{cell} = 0.4736\,V$

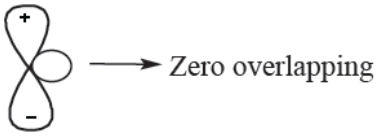
30. (4)

Based on intra-molecular aldol condensation Nucleohile attack on less hindered aldehyde to from stable six membered ring

31. (1)



Zero overlapping is present due to improper orientation

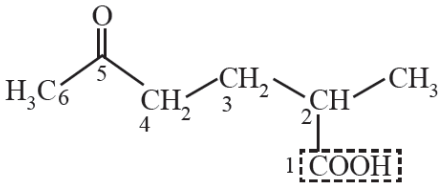


32. (2)

Aqueous KOH causes substitution, while the alc. KOH causes elimination reaction.

33. (2)

In the given compound, the numbering will start from —COOH group. Double bond O present at 5th position will named as 'oxo' in prefix.



IUPAC name of the above compound is:
2–Methyl–5–oxohexanoic acid

34. (3)

$P^\circ_A = 7 \times 10^3$ $P^\circ_B = 12 \times 10^3$

$X_A = 0.4$ $X_B = 0.60$

$P_T = X_A P^\circ_A + X_A P^\circ_B = 0.60$

$= 7 \times 10^3 \times 0.4 + 12 \times 10^3 \times 0.60$

$= 2.8 \times 10^3 + 7.2 \times 10^3$

$P_T = 10 \times 10^3$

$y_A = \frac{P_A}{P_T}, y_B = \frac{P_B}{P_T}$

$y_A = \frac{2.8 \times 10^3}{10 \times 10^3}, y_B = \frac{7.2 \times 10^3}{10 \times 10^3}$

$y_A = 0.28, y_B = 0.72$

35. (4)

Sodium react with C, N, S and X to form following compounds.

$Na + C + N \rightarrow NaCN$

$2Na + S \xrightarrow{\Delta} Na_2S$

$Na + X \rightarrow NaX$

Test for nitrogen

$6NaCN + FeSO_4 \rightarrow Na_4\left[Fe(CN)_6\right] + Na_2SO_4$

$Na_4\left[Fe(CN)_6\right] + Fe^{3+} \rightarrow Fe_4\left[Fe(CN)_6\right]_3$

(Prussian blue)

Sulphur detection by Sodium nitroprusside

$Na_2\left[Fe(CN)_5NO\right] + Na_2S \rightarrow Na_4\left[Fe(CN)_5NOS\right]$

[Purple]

Phosphorus detection by ammonium molybdate

$Na_3PO_4 + 3HNO_3 \rightarrow H_3PO_4 + 3NaNO_3$

$H_3PO_4 + 12\left(NH_4\right)_2 MoO_4 + 21HNO_3 \rightarrow$

$\left(NH_4\right)_3 PO_4 \cdot 12MoO_3 + 21NH_4NO_3 + 12H_2O$

(canary yellow)

Halogen give specific coloured ppt with $AgNO_3(aq)$

$NaCl + AgNO_3(aq) \rightarrow AgCl + NaNO_3$

(White)

$NaBr + AgNO_3(aq) \rightarrow AgBr + NaNO_3$

(Paleyellow)

$NaI + AgNO_3(aq) \rightarrow AgI + NaNO_3$

(Yellow)

36. (1)

$\left[Fe\left(H_2O\right)_6\right]Cl_2, Fe^{2+} \rightarrow 3d^6 \rightarrow \left(t_{2g}\right)^4\left(e_g\right)^2$

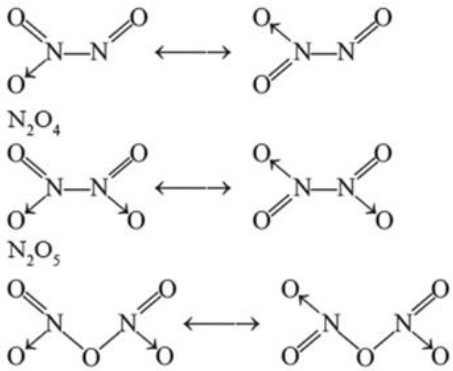
C.F.S.E. $= 4 \times \left(-0.4\Delta_o\right) + 2 \times 0.6\Delta_o = -0.4\Delta_o$

$K_2\left[NiCl_4\right], Ni^{2+} \rightarrow 3d^8 \rightarrow (e)^4\left(t_2\right)^4$

C.F.S.E. $= 4 \times \left(-0.6\Delta_t\right) + 4 \times \left(0.4\Delta_t\right) = -0.8\Delta_t$

37. (3)

The structure of various given oxides of N are:-



Hence, N_2O, N_2O_3 and N_2O_4 has N – N bond.

38. (2)

Higher the electron density on benzene ring, Higher the reactivity towards electrophileic substitution.

39. (4)

The empirical formula for the compound A can be

calculated as:

	Mass	Moles $= \frac{\text{Weight}}{\text{Molar Mass}}$	Simplest Ratio
C	74g	$\frac{74}{12} = 6.17$	$\frac{6.17}{1.24} = 4.9 \approx 5$
H	8.7g	$\frac{8.7}{1} = 8.7$	$\frac{8.7}{1.24} = 7$
N	17.3g	$\frac{17.3}{14} = 1.24$	$\frac{1.24}{1.24} = 1$

Hence, empirical formula is C_5H_7N and empirical mass is 81 g/mol

Given: Molecular mass = 162 g/mol

$$\Rightarrow n = \frac{\text{Molecular Mass}}{\text{EmpiricalMass}} = \frac{162}{81} = 2$$

Molecular formula = n× Empirical formula = $C_{10}H_{14}N_2$

40. (4)

In CrO_2Cl_2 oxidation state of Cr is +6

No. of electrons present in various ions.

Cr (VI) = $[Ar]^{18}\ 3d^0$

(1) $Ti(III) = [Ar]^{18}3d^1$

(2) $Fe(III) = [Ar]^{18}3d^5$

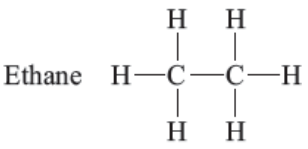
(3) $V(IV) = [Ar]^{18}3d^1$

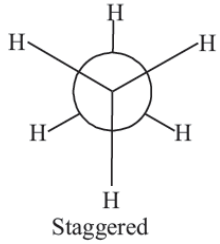
(4) $Mn(VII) = [Ar]^{18}3d^0$

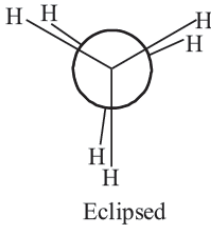
Hence, Cr (VI) and Mn (VII) have same d^0 configuration.

41. (4)

Rotamers or conformers arises when there is a free rotation along σ - bond.



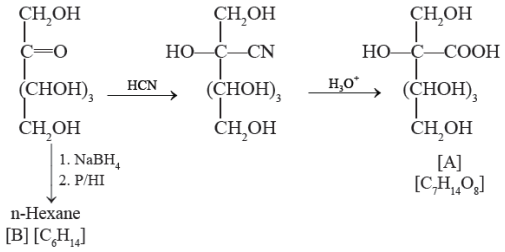




42. (1)

Due to presence of two hydrogen atoms, in primary amines, inter molecular association is more pronounced.

43. (1)



44. (3)

On moving down a group, Electronegativity decreases and atomic radius increases for representative elements.

45. (2)

3° alcohols react fastest with $ZnCl_2$/conc.HCl due to formation of 3° carbocation and 2–methyl propan–2–ol is the only 3° alcohol.

46. (15)

$[A]_t = [A]_0e^{-kt}$

For A: Half-life for the first order reaction $[A]_t = y$, $[A_0] = x$.

$$k = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{15\,\text{min}}$$

$$= xe^{-\left(\frac{\ln 2}{15}\right)t}$$

$$\text{For B}: [B]_t = [B]_0\, e^{-kt}$$

$$\text{Let } [B]_t = y; [B]_0 = 4x;\ k = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{15\,\text{min}}$$

$$y = 4xe^{-\left(\frac{\ln 2}{15}\right)t}$$

$$\Rightarrow xe^{-\left(\frac{\ln 2}{5}\right)t} = 4xe^{-\left(-\frac{\ln 2}{15}\right)t}$$

$$e^{t\left(\frac{\ln 2}{5} - \frac{\ln 2}{15}\right)} = 4; t \times \left[\frac{\ln 2}{5} - \frac{\ln 2}{15}\right] = \ln 4$$

$$t \times \ln 2\left[\frac{1}{5} - \frac{1}{15}\right] = 2\ln 2$$

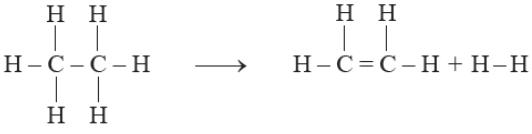
$$t = 15\,\text{min}$$

47. (128)

$C_2H_6 \rightarrow C_2H_4 + H_2$

When bond is formed, Energy is released

When bond is broken, Energy is consumed



$\Delta H_f = [\Delta H_{C-C} + 6\Delta H_{C-H}] - \Delta H_{C=C} - 4\Delta H_{C-H} - \Delta H_{H-H}$

$= \Delta H_{C-C} + 2\Delta H_{C-H} - \Delta H_{C=C} - \Delta H_{H-H}$

Given: $\Delta H_{C-C} = 347\,kJ/mol$

$\Delta H_{C-C} = 611 kJ/mol$

$\Delta H_{C-H} = 414 kJ/mol$

$\Delta H_{H-H} = 436 kJ/mol$

$= 347 + 2\times 414 - 611 - 436$

$= 128 kJ/mol$

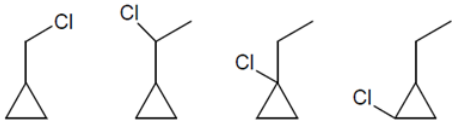
48. (25)

$MnO_4^- + 8H + 5e^- \longrightarrow Mn^{2+} + 4H_2O \quad E^\circ = 1.51V$

$\because$ 1 mole of MnO_4^- required 5 moles of electrons or 5F electricity

5 moles required = 25F

49. (4)



50. (25)

	A	+	B	$\rightleftharpoons$	2C
Initially	1		1		1
At eq.	$1-x$		$1-x$		$(1+2x)$

$\therefore 100 = \dfrac{(1+2x)^2}{(1-x)(1-x)}$

$\left(\dfrac{1+2x}{1-x}\right) = 10$

$\therefore x = 3/4,$

Now,

$\therefore [C] = (1+2x) = 1 + 2\times\dfrac{3}{4} = \left(1+\dfrac{3}{2}\right)M$

$= 2.5M = 25\times 10^{-1} M$

SECTION-III (MATHEMATICS)

51. (4)

$7x^2 - 3x - 2 = 0 \Rightarrow \alpha+\beta = \dfrac{3}{7}, \alpha\beta = \dfrac{-2}{7}$

Now, $\dfrac{\alpha}{1-\alpha^2} + \dfrac{\beta}{1-\beta^2}$

$= \dfrac{\alpha - \alpha\beta(\alpha+\beta) + \beta}{1-\left(\alpha^2+\beta^2\right) + (\alpha\beta)^2}$

$= \dfrac{(\alpha+\beta) - \alpha\beta(\alpha+\beta)}{1-(\alpha+\beta)^2 + 2\alpha\beta + (\alpha\beta)^2}$

$= \dfrac{\frac{3}{7} + \frac{2}{7}\times\frac{3}{7}}{1 - \frac{9}{49} + 2\times\frac{-2}{7} + \frac{4}{49}} = \dfrac{21+6}{49-9-28+4} = \dfrac{27}{16}$

52. (3)

$A^T = \alpha A + I$

Take transpose of matrix,

$\Rightarrow A = \alpha A^T + I$

$\Rightarrow A = \alpha(\alpha A + I) + I \qquad \left[\because A^T = \alpha A + T\right]$

$\Rightarrow A = \alpha^2 A + (\alpha+1)I$

$\Rightarrow A\left(1-\alpha^2\right) = (\alpha+1)I$

$\Rightarrow A = \dfrac{I}{1-\alpha} \qquad [\because \alpha \neq 1] \quad \ldots (i)$

$\Rightarrow |A| = \dfrac{1}{(1-\alpha)^2} \qquad \ldots. (ii)$

Now det $\left(A^2 - A\right) = 4$

$\Rightarrow |A||A-I| = 4 \qquad \ldots. (iii)$

Now, $A - I = \dfrac{I}{1-\alpha} - I = \dfrac{\alpha}{1-\alpha} I$

$\Rightarrow |A-I| = \left(\dfrac{\alpha}{1-\alpha}\right)^2 \qquad \ldots..(iv)$

Using equation (iii)

$\Rightarrow \dfrac{1}{(1-\alpha)^2}\dfrac{\alpha^2}{(1-\alpha)^2} = 4$

$\Rightarrow \dfrac{\alpha}{(1-\alpha)^2} = \pm 2 \Rightarrow 2(1-\alpha)^2 = \pm\alpha$

Case-I: $2(1-\alpha)^2 = \alpha$ Case-II: $2(1-\alpha)^2 = -\alpha$

$\Rightarrow 2\alpha^2 - 5\alpha + 2 = 0 <^{\alpha_1}_{\alpha_2}$ $\Rightarrow 2\alpha^3 - 3\alpha + 2 = 0$

$\alpha_1 + \alpha_2 = \dfrac{5}{2}$ $\alpha \notin$ Real values

Sum of value of $\alpha = \dfrac{5}{2}$

53. (2)

$\sin\left(\dfrac{\pi}{22}\right) = \cos\left(\dfrac{\pi}{2} - \dfrac{\pi}{22}\right) = \cos\dfrac{5\pi}{11} = -\cos\dfrac{16\pi}{11}$

$\sin\dfrac{3\pi}{22} = \cos\dfrac{4\pi}{11}, \sin\dfrac{5\pi}{22} = \cos\dfrac{3\pi}{11} = -\cos\dfrac{8\pi}{11}$

$\sin\dfrac{7\pi}{22} = \cos\dfrac{2\pi}{11}, \sin\dfrac{9\pi}{11} = \cos\dfrac{\pi}{11}$

Now, $2\sin\dfrac{\pi}{22}\cdot\sin\dfrac{3\pi}{22}\cdot\sin\dfrac{5\pi}{22}\cdot\sin\dfrac{7\pi}{22}\cdot\sin\dfrac{9\pi}{22}$

$2\cos\dfrac{\pi}{11}\cdot\cos\dfrac{2\pi}{11}\cdot\cos\dfrac{4\pi}{11}\cdot\cos\dfrac{8\pi}{11}\cos\dfrac{16\pi}{11}$

$\dfrac{2\times\sin 2^5\dfrac{\pi}{11}}{2^5\sin\dfrac{\pi}{11}} = \dfrac{1}{16}\left(\because \sin 2^5\dfrac{\pi}{11} = \sin\dfrac{\pi}{11}\right)$

54. (1)

$$\left(-2-\frac{i}{3}\right)^3=\left(\frac{x+iy}{27}\right)$$

$$=(-1)^3\left(2^3+\frac{i^3}{27}+3(2)\frac{i^2}{9}+3(2)^2\cdot\frac{i}{3}\right)=\frac{x+iy}{27}$$

$$-\left[8-\frac{i}{27}-\frac{2}{3}+4i\right]=\frac{x+iy}{27}$$

Comparing real and imaginary part

$$\Rightarrow \frac{x}{27}=-8+\frac{2}{3} \text{ and } \frac{y}{27}=\frac{1}{27}-4$$

So, $\frac{y-x}{27}=\frac{1}{27}-4+8-\frac{2}{3}=\frac{1+27\times4-18}{27}=\frac{91}{27}$

$$\Rightarrow y-x=91$$

55. (3)

$$\left.\begin{matrix}2x+y=4\\2x+6y=14\end{matrix}\right\}\Rightarrow y=2, x=1$$

$B(1, 2)$

Let $C(k, 4-2k)$

Now $AB^2=AC^2$

$$5^2+(-1)^2=(6-k)^2+(-3+2k)^2$$

$$\Rightarrow 5k^2-24k+19=0$$

$$\Rightarrow (5k-19)(k-1)=0$$

$$\Rightarrow k=\frac{19}{5}$$

$$C\left(\frac{19}{5},-\frac{18}{5}\right)$$

Centroid (a, b)

$$\alpha=\frac{6+1+\frac{19}{5}}{3}=\frac{18}{5}$$

$\beta = (1 + 2 – 18/5)/3 = – 1$

Therefore 15 ($\alpha + \beta$)

$$=15\left(\frac{17}{5}\right)=51$$

56. (4)

Equation of OP is $\frac{x}{3}=\frac{y}{4}=\frac{z}{5}$

$a_1 = (0, 0, 0)$ $a_2 = (3, 0, 5)$

$b_1 = (3, 4, 5)$ $b_2 = (0, 0, 1)$

Equation of edge parallel to z axis

$$\frac{x-3}{0}=\frac{y-0}{0}=\frac{z-5}{1}$$

$$\text{S.D}=\frac{(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_2\times\vec{b}_1)}{|\vec{b}_2\times\vec{b}_1|}$$

$$\frac{\begin{vmatrix}3&0&5\\3&4&5\\0&0&1\end{vmatrix}}{\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&4&5\\0&0&1\end{vmatrix}}=\frac{3(4)}{|4\hat{i}-3\hat{j}|}=\frac{12}{5}$$

57. (1)

$$\text{Median } =35=\frac{\left(\frac{n}{2}\right)th+\left(\frac{n}{2}+1\right)th}{2},\ n = \text{even}$$

$$\frac{34+x}{2}=35$$

$$\Rightarrow x=36$$

$$\text{Mean } =42$$

$$=\frac{10+22+26+29+34+36+42+67+70+y}{10}$$

$$\Rightarrow 420-336=y\Rightarrow y=84$$

$$\Rightarrow \frac{y}{x}=\frac{84}{36}=\frac{7}{3}$$

58. (4)

Number of terms in 11th set

1,3,5,7……..

A11 = 1 + (10)(2) = 21

first term of 11t set will be last term of 10th set + 3

So Sum of term upto 10th set

$$S_{10}=\frac{10}{2}[2\times1+9\times2]=100$$

So, last term of 10th set is 300.

Therefore, 11th set first term = 303

Then sum of all terms of 11th set

$$S=\frac{21}{2}[2\times303+20\times3]$$

= 6993

59. (4)

Eliminating 't' from the given equations we get equation $x^2 = 12y$, which is a parabola.

Focus is $S(0, 3)$.

Let $A\left(2t,\frac{t^2}{3}\right)$, $B(0,\ p)$

Now, $SA\perp BA$

$$\therefore mSA\cdot mBA=-1$$

$$\Rightarrow \frac{\left(3-\frac{t^2}{3}\right)}{(0-2t)}\cdot\frac{\left(p-\frac{t^2}{3}\right)}{(0-2t)}=-1$$

$$\Rightarrow 3p=\frac{27t^2+t^4}{t^2-9}$$

$$\text{centroad } \frac{1}{3}\left[3+\frac{t^2}{3}+p\right]-\text{ ordinate}$$

$$\Rightarrow \lim_{t\to1}k=\lim_{t\to1}\frac{1}{9}\left(9+t^2+\frac{27t^2+t^4}{\left(t^2-9\right)}\right)=\frac{13}{18}$$

60. **(3)**

$$f(x)=\begin{cases}\left|4x^2-8x+5\right|, & x\le\frac{1}{4}\text{ or }x\ge\frac{1}{2}\\ \left[4x^2-8x+5\right], & \frac{1}{4}<x<\frac{1}{2}\end{cases}$$

Now, $4x^2 - 8x + 5 > 0$ for all real x.

Point of minima of $4x^2 - 8x + 5$ is $x = 1$.

So, $g(x) = 4x^2 - 8x + 5$ decreases in (1/4, 1/2)

$g(1/4) = 3 + 1/4$, $g(1/2) = 2$

So, $g(x)$ will be discontinuous for $4x^2 - 8x + 5 = 3$ or $2x^2 - 4x + 1 = 0$

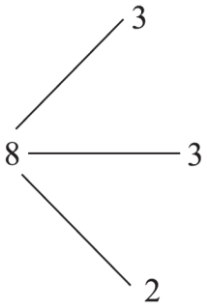
$$\therefore x=1-\frac{1}{\sqrt{2}}\in\left(\frac{1}{4},\frac{1}{2}\right)$$

$$\therefore f(x)=\begin{cases}4x^2-8x+5, & x\le\frac{1}{4}\\ 3, & \frac{1}{4}<x<1-\frac{1}{\sqrt{2}}\\ 2, & 1-\frac{1}{\sqrt{2}}\le x<\frac{1}{2}\\ 4x^2-8x+5, & x\ge\frac{1}{2}\end{cases}$$

So, $f(x)$ is discontinuous and hence non-differentiable at $x=\frac{1}{4},1-\frac{1}{\sqrt{2}},\frac{1}{2}$.

61. **(2)**

Group 8 person such that



Number of ways $=\frac{8!}{3!3!2!2!}\times 3!$

$=\frac{8\times7\times6\times5\times4}{4}$

= 56 × 30

= 1680

62. **(2)**

$$f(x)=\frac{\text{cosec}^{-1}x}{\sqrt{x-[x]}}$$

$\therefore \text{cosec}^{-1}x$ is defined on $(-\infty,-1]\cup[1,\infty)$

and $x-[x]\neq 0$

$\{x\}\neq 0, x\notin I$

Hence, $f(x)$ is defined for all non integer except in the interval [–1,1].

63. **(2)**

Here, $e^{3\log_e 2x}=8x^3$, similarly for other terms.

$$I=\int\frac{8x^3+20x^2}{x^4+5x^3-7x^2}dx=4\int\frac{2x+5}{x^2+5x-7}dx$$

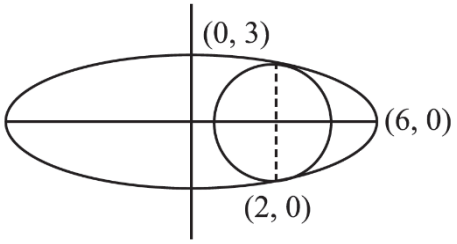
$$=4\ln\left|x^2+5x-7\right|+c$$

64. **(3)**

$$\frac{x^2}{36}+\frac{y^2}{9}=1$$

Let circle equation with r is: $(x-2)^2+y^2=r^2$

solving with ellipse, we get



$$(x-2)^2+\frac{36-x^2}{4}=r^2$$

$$3x^2-16x+52-4r^2=0$$

$D = 0$

$$\Rightarrow 16^2-4\times3\times\left(52-4r^2\right)=0\Rightarrow 4r^2=\frac{92}{3}$$

Hence, $12r^2=92$.

65. **(3)**

Four cases = (0, 0) & (1, 1) are roots

Total cases = (0, 0), (0, 1), (1, 1) & (ω, ω^2)

$$=\frac{2}{4}=\frac{1}{2}$$

66. **(3)**

Given equation of hyperbola is

$$\frac{x^2}{6/k}-\frac{y^2}{6}=1 \qquad \text{..... (i)}$$

Take $e^2=1+\frac{b^2}{a^2}$

$$\Rightarrow e^2=1+\frac{6}{6/k}$$

$$\Rightarrow e=\sqrt{1+k}$$

Here, $a=\sqrt{\frac{6}{k}}$

Equation of directrix x $x=\frac{a}{e}\Rightarrow x=\sqrt{\frac{6}{k(k+1)}}$

$$\Rightarrow k^2+k-6=0$$

$\Rightarrow k = 2, -3$ (Rejected)

From eq. (i) we get $2x^2 - y^2 = 6$

Satisfy the coordinate $(\sqrt{5},-2)$ then it holds the required equation.

67. (1)

Given, $\frac{dy}{dx}=\frac{y}{x}+\frac{\sqrt{y^2+16x^2}}{x}$

Put y $V.$ x in the above eq^n and

$\frac{dy}{dx}=V+x\cdot\frac{dV}{dx}$, We get

$\Rightarrow V+x\frac{dV}{dx}=V+\sqrt{V^2+16}$

$\Rightarrow x\frac{dV}{dx}=\sqrt{V^2+16}$

apply variable separable method, we get

$\Rightarrow\int\frac{dV}{\sqrt{V^2+16}}=\int\frac{dx}{x}+\ell nC$

$\Rightarrow \ell n\left|V+\sqrt{V^2+16}\right|=\ell nCx$

$\Rightarrow\frac{y}{x}+\frac{\sqrt{y^2+16x^2}}{x}=Cx$

Given $y(1) = 3 \Rightarrow C = 8$

Now at $x = 2$

$\Rightarrow y+\sqrt{y^2+64}=32$

$\Rightarrow y^2+64=(32-y)^2$

$=y+64=y^2-64y+32^2$

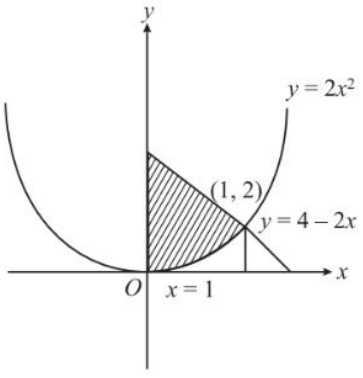
$\Rightarrow y=15$

68. (3)

$\left\{(x,y)\in R\times R, x\geq 0, 2x^2\leq y\leq 4-2x\right\}$

$\therefore$ Required area $\int_0^1\left(4-2x-2x^2\right)dx$

$=\left(4x-x^2-\frac{2}{3}x^3\right)\Big|_0^1=\frac{7}{3}$



69. (3)

Given r 10, $R = 10 + h$ and $\frac{dV}{dt}=-50$

$V=\frac{4}{3}\pi\left((R)^3-r^3\right)$

$\Rightarrow V=\frac{4}{3}\pi\left((10+h)^3-10^3\right)$

Diff. w.r.t. time.

$\frac{dV}{dt}=4\pi(10+h)^2\frac{dh}{dt}$

$\Rightarrow -50=4\pi(10+5)^2\frac{dh}{dt}\Rightarrow\frac{dh}{dt}=-\frac{1}{18\pi}\frac{\text{cm}}{\text{min}}$

70. (4)

Use $f(x)=\begin{cases}-x, & \text{if} \quad -z<x<-1\\ x+2, & \text{if} \quad -1<x<0\\ 3x+2, & \text{if} \quad 0<x<1\\ x+4, & \text{if} \quad 1<x<2\end{cases}$

$f'\left(\frac{-3}{2}\right)=-1, f'\left(\frac{-1}{2}\right)=1, f'\left(\frac{1}{2}\right)=3, f'\left(\frac{3}{2}\right)=1$

$S_1 f'\left(\frac{-3}{2}\right)=+f'\left(\frac{-1}{2}\right)+f'\left(\frac{1}{2}\right)+f'\left(\frac{3}{2}\right)$

$=-1+1+3+1=4$

And S_2:

$\int_{-2}^{2}f(x)dx=\int_{-2}^{-1}f(x)dx+\int_{-1}^{0}f(x)dx+\int_{0}^{1}f(x)dx+\int_{1}^{2}f(x)dx$

$=\int_{-2}^{-1}(-x)dx+\int_{-1}^{0}(x+2)dx$

$+\int_{0}^{1}(3x+2)dx+\int_{1}^{2}(x+4)dx$

$= 12$

71. (7)

$(2023)^{2023}$

$=(2030-7)^{2023}=(35K-7)^{2023}$

$={}^{2023}C_0(35K)^{2023}(-7)^0+{}^{2023}C_1(35K)^{2022}(-7)+$

$\ldots\ldots+\ldots\ldots+{}^{2023}C_{2023}(-7)^{2023}$

$=35N-7^{2023}$

Now, $-7^{2023}=-7\times7^{2022}=-7\left(7^2\right)^{1011}$

$=-7(50-1)^{1011}$

$=-7\left({}^{1011}C_0 50^{1011}-{}^{1011}C_1(50)^{1010}+\ldots\cdot{}^{1011}C_{1011}\right)$

$=-7(5\lambda-1)$

$=-35\lambda+7$

$\therefore$ When $(2023)^{2023}$ is divided by 35 remainder is 7

72. **(2)**

The circle $x^2+y^2+6x+8y+16=0$ has centre

the circle

$x^2+y^2+2(3-\sqrt{3})x+2(4-\sqrt{6})y$

$=k+6\sqrt{3}+8\sqrt{6}$

has centre $(\sqrt{3}-3,\sqrt{6}-4)$ and radius $\sqrt{k+34}$

$\because$ These two circles touch internally hence

$C_1C_2=|r_1-r_2|$

$\sqrt{3+6}=|\sqrt{k+34}-3|$

Here, k = 2 is only possible ($\because$ k > 0)

73. **(18)**

$A=\{1,2,3,\ldots..10\}$

$B=\{0,1,2,3,4\}$

$R=\left\{(a,b)\in A\times A:2(a-b)^2+3(a-b)\in B\right\}$

$\therefore 2(a-b)^2+3(a-b)=(a-b)(2(a-b)+3)$

$\Rightarrow a=b$ or $a-b=-2$

When a = b $\Rightarrow$ 10 order pairs

When $a=b$ $\Rightarrow$ 8 order pairs

Total = 18

74. **(36)**

$|\vec{a}|=\sqrt{14},|\vec{b}|=\sqrt{6}$ and $|\vec{a}\times\vec{b}|=\sqrt{48}$

we know $|\vec{a}\times\vec{b}|^2+|\vec{a}\cdot\vec{b}|^2=|\vec{a}|^2\times|\vec{b}|^2$

$\Rightarrow(a\cdot\vec{b})^2=84-48=36$

75. **(120)**

$\left(1+x+x^2+x^3\right)^6=\left(\frac{1-x^4}{1-x}\right)^6$

Coefficient of $x^4\left(\frac{1-x^4}{1-x}\right)^6$ = coefficient of x^4 in

$\left(1-6x^4\right)(1-x)^{-6}$ = coefficient of x^4 in

$\left(1-6x^4\right)\left[1+{}^6C_1x+{}^7C_2x^2+\ldots\right]$

$={}^9C_4-6.1=126-6=120$



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if